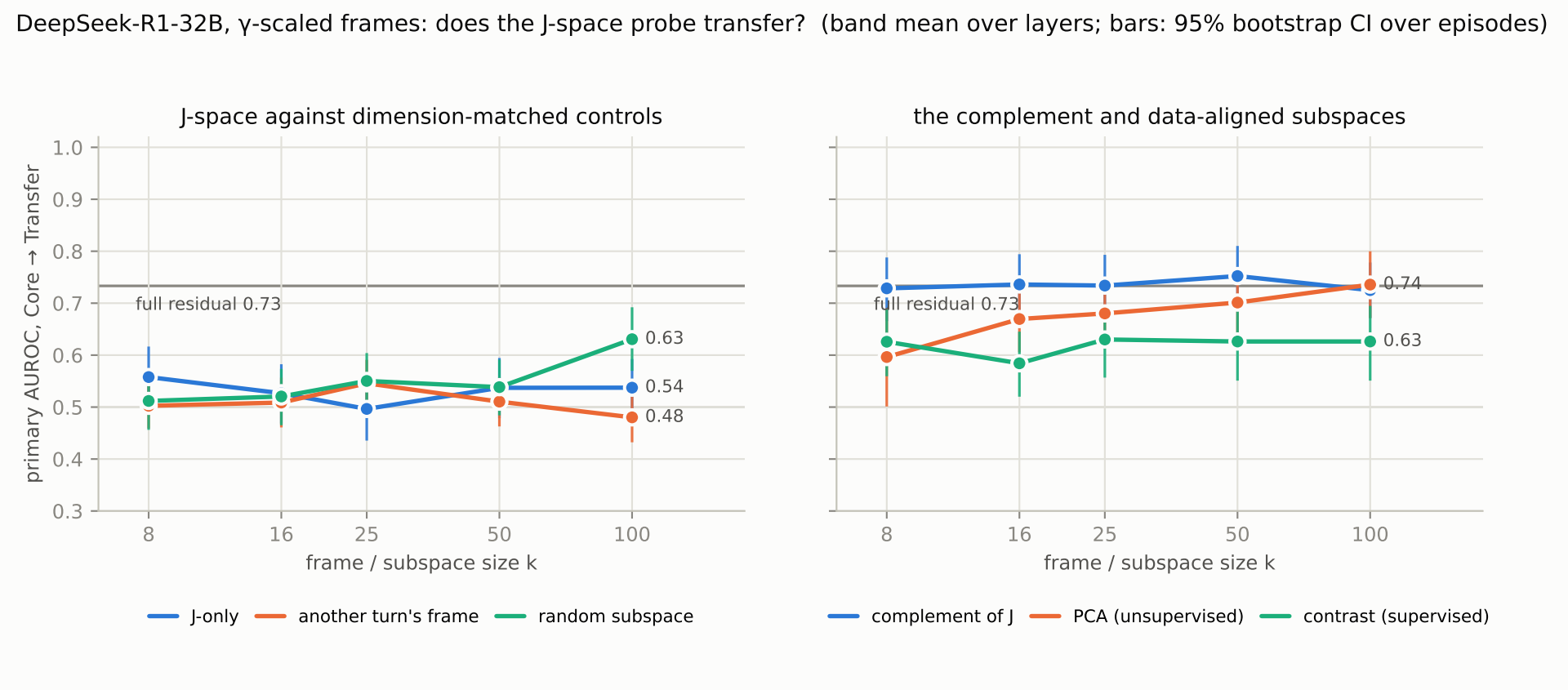
DeepSeek-R1-32B, γ-scaled frames: does the J-space probe transfer? (band mean over layers; bars: 95% bootstrap CI over episodes)
J-space against dimension-matched controls
the complement and data-aligned subspaces
primary AUROC, Core → Transfer
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
full residual 0.73
0.63
0.54
0.48
0.74
8
16
25
50
100
frame / subspace size k
J-only
another turn's frame
random subspace
complement of J
PCA (unsupervised)
contrast (supervised)